

Experiment 1: Objective Definition Document

Course: CSE4001 — Applications of Machine Learning in Industry | Target: Malignant (1) vs. Benign (0)

1. Problem Statement & Clinical Context

A diagnostic pathology laboratory processes high volumes of Fine-Needle Aspirate (FNA) biopsy samples for suspected breast cancer. Manual microscopic examination of nuclear morphological features is labor-intensive and subject to inter-observer variability. The objective is to formulate an automated second-opinion machine learning triage system to analyze digitized morphological features of cell nuclei and automatically flag likely-malignant tumours for priority pathologist review.

2. Machine Learning Formulation

Dimension	Specification
ML Experiment Type	Supervised Binary Classification
Input Features	30 continuous nuclear morphological measurements (mean, SE, worst of 10 geometric features)
Target Variable	Diagnosis: 0 = Benign, 1 = Malignant (Binary Flag)
Primary Metric	Malignant Class Recall / Sensitivity ($\geq 98\%$) to minimize missed malignancies
Secondary Metrics	Precision ($\geq 90\%$), ROC-AUC (≥ 0.98), F1-Score, False Negative Rate (FNR)
Deployment Latency	< 200 ms per sample via RESTful FastAPI scoring service

3. Clinical Success Metrics & Rationale

In clinical oncology screening, a **False Negative** (classifying a malignant tumor as benign) carries fatal risk, as vital cancer therapy is delayed. Conversely, a **False Positive** results in routine pathologist review. Therefore, the ML optimization objective strictly prioritizes Malignant-class Recall over raw overall accuracy.

4. Stakeholder Analysis

- **Pathologists & Cytotechnologists:** Clinical end-users receiving prioritized alerts and confidence scores.
- **Patients:** Benefit from accelerated turnaround times and reduced diagnostic error.
- **Laboratory Directors:** Ensure clinical QA, HIPAA/GDPR regulatory compliance, and auditability.
- **ML Engineering Team:** Responsible for training, validating, deploying, and monitoring the predictive pipeline.

5. Operational Constraints & Governance

- **Second-Opinion Decision Support:** The system acts as assistive decision support and does not replace certified pathologist review.
- **Class Imbalance Management:** The ~63% benign to 37% malignant ratio requires threshold tuning and loss weighting.
- **Zero Data Leakage:** Preprocessing statistics must strictly derive from training partitions.